



Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences

Department of Computer Science & Information Technology

Mid Term Examinations (Fall 2022)

Course Code with Title	CS-212T: Data Structures and Algorithms		Program	Bachelor of Computer Science (BSCS)
Instructor	Adeel Saeed / Muhammad Fahim		Semester	3 rd
Start date & Time	Dec 05, 2022 at 11:30 AM	End Time	01:00 PM	
Maximum Marks	30			

IMPORTANT INSTRUCTION:

Read the following Instruction carefully:

- Attempt All Questions.

Q.1. (05)

Consider an array with values {1, 3, 9, 8, 2, 7, 5}, illustrate step by step representation of Selection Sort algorithm to sort the given array in ascending order. Also give Selection Sort algorithm or pseudocode.

Q.2. (05)

Given 5 elements queue Q (from front to back: 1, 3, 5, 7, 9), and an empty stack S, remove the elements one-by-one from Q and insert them into S, then remove them one-by-one from S and re-insert them into Q. What sequence does the queue Q (from front to back) contains?

Q.3. (05)

Given a Queue data structure that supports standard operations like enqueue() and dequeue(). The task is to implement a Stack data structure using only instances of Queue and Queue operations allowed on the instances.

Q.4. (05)

Consider Multidimensional array with real life scenario. Suppose you went to buy ticket on ticket counter, it may be bus ticket there are two lines mentioned. One is for male and second one is for female. As there is more than one line, so it may be called as multi dimension (2 dimension in this case). So if there are multiple rows and multiple columns, then it is called as Multi-dimension. Suppose there are 4 lines at bus ticket counter. Each person is standing at its position starting with [0][0] position and ending with [5][3] to collect tickets. Now represent such scenario in C++ using 2 Dimensional Array.



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Q.5. (05)
 For each of the following program fragments, give an analysis of the running time Big- Oh.

- (1) `sum = 0;` $\rightarrow 1$
`for( i = 0; i < n; i++ )` $\rightarrow n$
`sum++;` $\rightarrow n$ n^2
- (2) `sum = 0;`
`for( i = 0; i < n; i++ )` $\rightarrow n^2$
`for( j = 0; j < n; j++ )` $\rightarrow n^3$
`sum++;` $\rightarrow n^3$
- (3) `sum = 0;`
`for( i = 0; i < n; i++ )` $\rightarrow n^3$
`for( j = 0; j < n * n; j++ )` $\rightarrow n^3$
`sum++;`
- (4) `sum = 0;`
`for( i = 0; i < n; i++ )` $\rightarrow n^2$
`for( j = 0; j < i; j++ )`
`sum++;`
- (5) `sum = 0;`
`for( i = 0; i < n; i++ )` $\rightarrow n^3$
`for( j = 0; j < i * i; j++ )`
`for( k = 0; k < j; k++ )`
`sum++;`

Q.6. (05)
 Create a Student Record Management system using singly linked list that can perform the following operations (algorithm or pseudocode):

- Insert Student record
- Show Student record

The student record should contain the following items

- Name of Student
- Roll Number of Student